

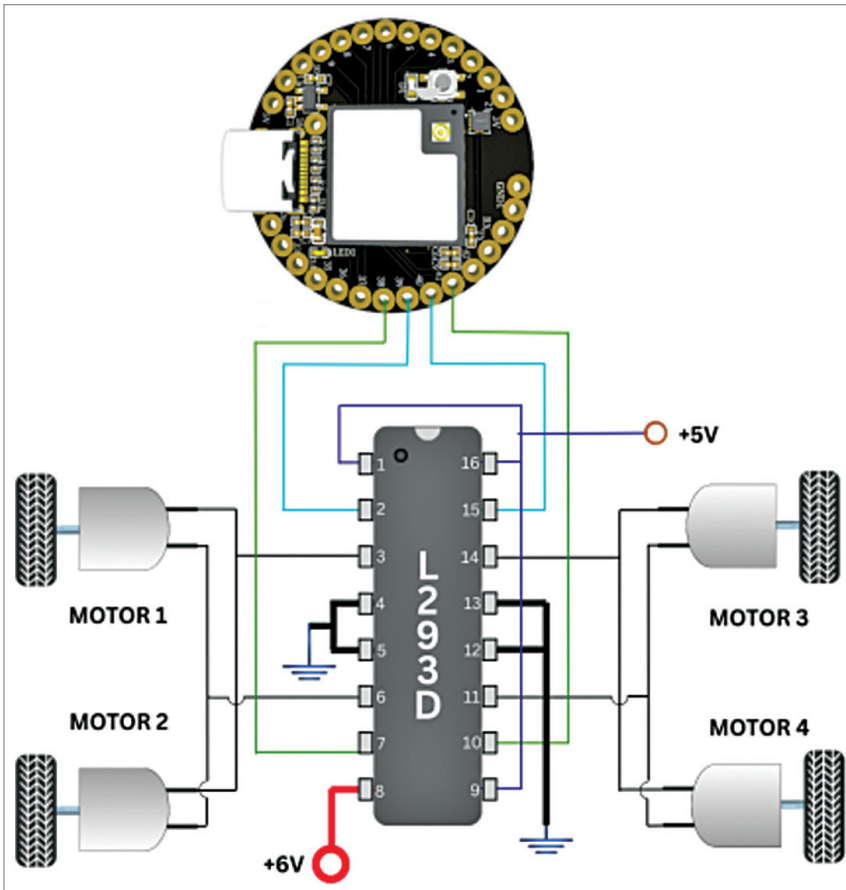


Fig. 5: Wi-Fi robot car circuit diagram

TABLE 2

ARDUINO IOT CLOUD DASHBOARD CONTROLS

Dashboard element	Function	IndusBoard Coin pins
Forward push button	Forward movement	39, 40
Backward push button	Backward movement	38, 41
Left push button	Left turn	38, 40
Right push button	Right turn	39, 41

- Open the generated thingProperties.h file and verify the Wi-Fi credentials and device key.

- In the callback functions, such as onForwardChange(), onBackwardChange(), onLeftChange(), and onRightChange(), add the necessary control statements to drive the motor control pins. Fig. 3 shows the relevant code snippet.

Now, to configure the dashboard, navigate to the dashboards tab and create a new dashboard. See Fig. 4 showing the IoT Cloud dashboard

Click Add→widgets and select the push button widget for the forward

variable; repeat the same for backward, left, and right by linking them with each other's widget, respectively.

Repeat this three more times

until there are four push button widgets on the screen.

Circuit and working

The circuit is built around an IndusBoard Coin (an ESP32-based controller), an L293D motor driver IC, four DC geared motors connected as two motor pairs, and a 5V-6V battery supply.

The IndusBoard Coin provides Wi-Fi connectivity and generates the control signals required to operate the motors through the L293D motor driver. The L293D functions as a dual H-bridge motor driver, allowing the

low-current GPIO pins of the IndusBoard Coin to control higher-current DC motors.

Digital pins 38, 39, 40, and 41 of the IndusBoard Coin are connected to the IN1, IN2, IN3, and IN4 input pins of the L293D, respectively. These inputs determine the direction of motor rotation. Fig. 5 shows the circuit diagram of the Wi-Fi-controlled robot car, which uses the IndusBoard Coin, an L293D motor driver, and four DC geared motors.

The ENA (pin 1) and ENB (pin 9) enable pins of the L293D are connected to the +5V supply, keeping both motor channels continuously enabled. The logic supply, VCC1 (pin 16), is connected to +5V, while the motor supply, VCC2 (pin 8), is powered from a 5V-6V battery source.

All ground terminals of the IndusBoard Coin, L293D, and battery are connected together to establish a common reference.

The left-side motor pair is connected across OUT1 (pin 3) and OUT2 (pin 6), while the right-side motor pair is connected across OUT3 (pin 11) and OUT4 (pin 14). Fig. 6 shows the interface between the IndusBoard Coin and the L293D motor driver for controlling the robot car's left and right drive motors.

When a control command is received from the Arduino IoT Cloud dashboard, the IndusBoard Coin processes the command and accordingly changes the logic levels on pins 38, 39, 40, and 41. These signals are applied to the L293D, which drives the motors in the required direction.

When the forward command is selected, both motor channels rotate forward, moving the robot forward. For backward movement, both motors rotate in reverse. During a left turn, the left motor is stopped or reversed while the right motor continues rotating, causing the robot to turn left. Similarly, during a right turn, the right motor is stopped or reversed while the left motor